ClimateIQ Smart AC Controller

Firmware structure & manual-command reference · FW 1.0.9 · ESP32 / FreeRTOS

1 · Overview — who runs what

The firmware is a set of independent **FreeRTOS tasks**. They never call each other directly — they coordinate through one global `sysData` struct, an event **queue**, a few **timers**, and **task notifications**. Crucially, *every* change to the AC is funnelled through one function, `executeACCommand()`, which sends the IR and ACKs.

Task	Responsibility
SensorsTask	Sole owner of the radar UART; presence/distance, HDC temp/humidity, the always-on IR remote listener, the button.
AutoTask	Drains the event queue & runs the AC state machine.
SchedTask	Evaluates the daily schedule each minute; offline failsafe.
NetworkTask	WiFi or GSM + MQTT; publishes telemetry, receives commands.
WebTask	Local AP dashboard: setup, IR learning, radar range/calibration.
OTATask	Idle until an `ota_update` job arrives; flash + trial rollback.
loop()	Watchdog, status LED, deferred reboot.

The 3 coordination channels

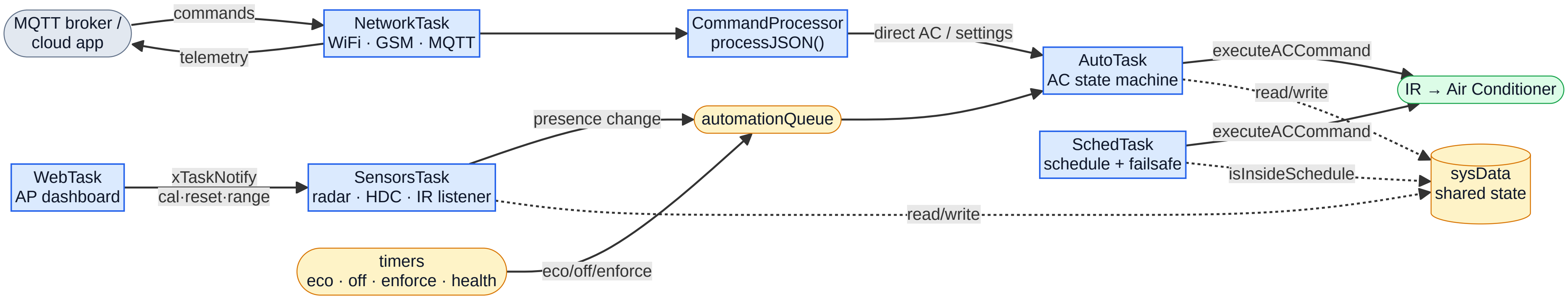
automationQueue — events (presence change, eco/off/enforce, manual override) wake *AutoTask*.

Timers — `eco` & `off` (room-empty countdowns), `enforce` (3-min re-assert), `health` (30s).

Task notifications — WebTask signals SensorsTask for radar calibrate / factory-reset / set-range / recover (it must touch the UART on the owning task).

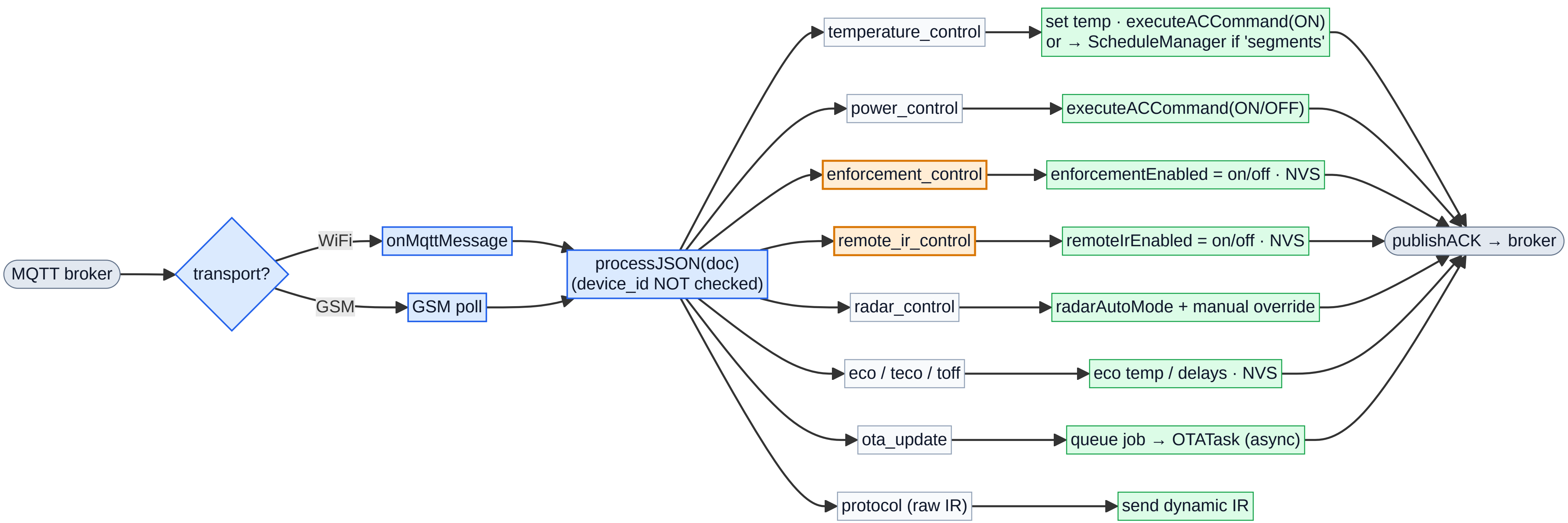
Your two new toggles live here: `enforcement_control` gates the enforce branch in *AutoTask*; `remote_ir_control` gates the listener in *SensorsTask*.

2 · Structure — tasks, IPC & data flow



Blue = FreeRTOS task · Amber = coordination primitive · Green = AC output · Grey = cloud.

3 · Manual commands — message → action



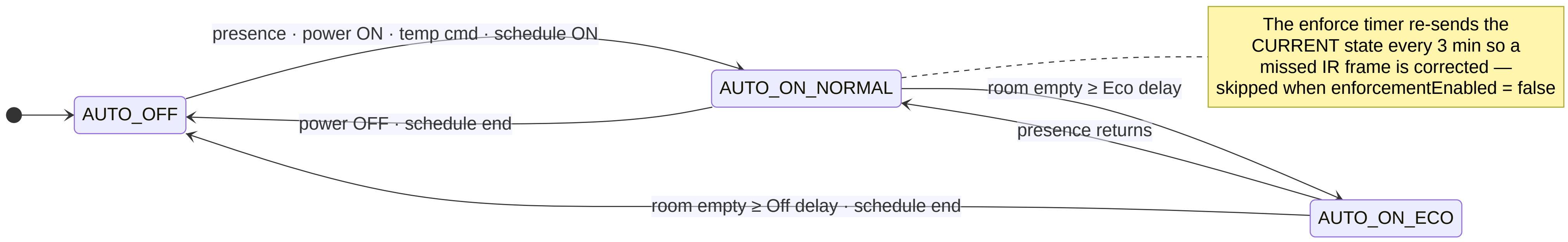
Amber-bordered boxes (`enforcement_control` , `remote_ir_control`) are the two new commands you added.

4 · Command reference

Command JSON	What it does	State / NVS key
`temperature_control` + `temperature_setting`	Set the AC setpoint now. Inside a schedule the stored normal temp is left intact (one-shot send). Outside schedule hours it sends, then immediately reverts to OFF.	`currentNormalTemp` · `normal_temp`
`temperature_control` + `segments`	A **schedule update** for a weekday → `ScheduleManager` (validate → sort → overlap-check → persist → apply if today).	per-day NVS blob
`power_control` + `power_status`	Turn AC ON (normal temp) or OFF. ON outside scheduled hours reverts to OFF.	`acAutoState`
`enforcement_control` + `enforcement_enabled` NEW	Enable/disable **only the periodic 3-min AC re-assertion**. The manual-remote revert is deliberately *not* affected.	`enforcementEnabled` · `enforce_en`
`remote_ir_control` + `remote_ir_enabled` NEW	Enable/disable the always-on AC-remote listener — detection, Remote-Activity logging, and the deviation revert.	`remoteIrEnabled` · `remote_ir_en`
`radar_control` + `radar`	Enable/disable radar presence automation; pins a manual override so the schedule won't flip it.	`radarAutoMode` · `radar_auto`
`eco` / `teco` / `toff`	Eco temperature, eco delay (min), off delay (min). Off delay is auto-bumped above eco delay.	eco_temp · eco_time · off_time
`ota_update` + `url` / `version`	Queue a server-pull firmware update; runs async in OTATask with trial-boot rollback.	—
`protocol` + `state` / `code`	Send a raw / dynamic IR frame via a named protocol.	—

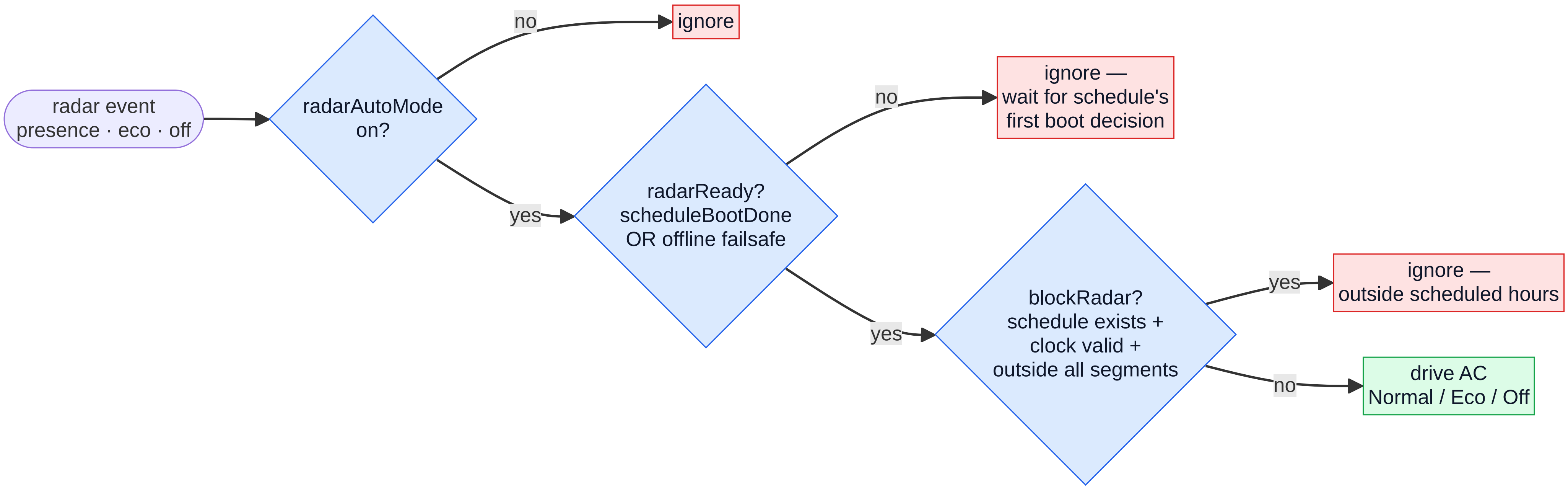
Reboot note for the two NEW toggles: their state is not sent in telemetry, so after a device restart the backend should re-send each command and wait for its `ack` (`action = enforcement` / `remote_ir`) to re-sync the dashboard buttons.

5 · AC state machine & what drives it



Three states only. Radar walks Normal→Eco→Off as a room empties; presence snaps it back to Normal.

6 · "Schedule is king" — when may radar act?



Offline failsafe: if the broker is unreachable AND the clock never synced for 60 s, SchedTask forces `radarAutoMode = true` and sets `isOfflineFailsafeActive` (solid magenta LED) so the room is still automated locally. When time/broker returns, the schedule resumes control. — Neither new toggle touches these gates, the schedule, or the eco/off countdowns.